

Task 0 — Agentic Incident Flow on Your PDI

EFFORT ~20 hours	WINDOW Deadline is Saturday, 5th of September, 2026 at 11:59 PM	MODE Individual work
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Overview

This is a real, small piece of automation. A support ticket is created in ServiceNow, your service decides what to do with it, and the answer appears back on that same ticket — automatically.

You do not need any ServiceNow experience. The asset pack includes a step-by-step guide, the trigger script, the exact data format, the knowledge articles, and three test tickets. The Gemini API key is free from Google AI Studio and needs no credit card.

The loop, step by step

The loop has four steps. Each one must work without manual help.

#	Step	What happens	Where
1	Ticket is created	Someone opens a new incident on your free ServiceNow developer instance (PDI).	ServiceNow
2	Ticket is sent out	A Business Rule fires by itself and sends the incident to your service as JSON.	ServiceNow → your service
3	Decision is made	Your service asks the Gemini LLM for one decision — respond, ask, or escalate — using only the five knowledge articles we give you.	Your Python service
4	Ticket is updated	Your service writes the result back onto the same ticket in ServiceNow.	Your service → ServiceNow

DONE MEANS

- The loop runs end to end with no manual steps.
- Bad input gets a clear error and repeated input is handled once.
- Someone else can clone your repository and run it from your README.

Technologies

ServiceNow PDI	Free developer instance. Setup is fully guided — no prior knowledge assumed.
Python 3.11+ · FastAPI	The webhook service you build.
Gemini LLM	The decision step. Free API key from Google AI Studio.
ngrok (or any free tunnel)	Gives your local service a public URL.
ServiceNow REST API	The write-back. Basic auth is fine for this task.
Git + GitHub	Your repository and commit history.

Asset pack

Everything below is inside `AI_Engineering_Task0_Assets.zip`, delivered with this document. There is no code skeleton — you build the service yourself.

<code>pdi_guide.md</code>	Step-by-step ServiceNow guide: request the PDI, expose your service with ngrok, create the Business Rule, test the trigger, and write results back — with the exact fields per decision.
<code>business_rule.js</code>	The ready Business Rule script. Paste it into the PDI and replace one URL.
<code>payload_contract.json</code>	The exact JSON your service will receive: <code>incident_sys_id</code> , <code>number</code> , <code>short_description</code> , <code>description</code> , <code>priority</code> — with an example.
<code>kb_articles.json</code>	The five knowledge articles. Your Gemini prompt must use these and nothing else.
<code>test_incidents.json</code>	Three test tickets with the decision each one must produce: a clear printer problem → respond; a vague email problem → ask; a leave request → escalate.

Functional requirements

FR1	Event capture	Set up your free PDI and create the Business Rule from the guide, so every newly created incident is sent to your service automatically.
FR2	Webhook service	Build a FastAPI service with a POST /webhook endpoint. It checks the payload and answers quickly with 202. The slow work runs in the background.
FR3	Gemini decision	Send the ticket text plus the five knowledge articles to the Gemini API. Ask for JSON with exactly two fields: a decision (respond, ask, or escalate) and a short message. Gemini must answer only from the given articles — if none of them fits, the decision is escalated.
FR4	Write-back	Update the same incident on the PDI through the ServiceNow REST API. respond → write the solution and resolve the ticket. ask → add a clarifying question as a customer-visible comment. escalate → add a work note saying it was escalated.
FR5	No double processing	If the same incident arrives twice, process it once. An in-memory guard is acceptable.
FR6	Test tickets	The three tickets in test_incidents.json must each produce their expected decision.

Non-functional requirements

NFR1	Fast answer	/webhook replies within about 2 seconds. The Gemini call and the write-back happen after that reply.
NFR2	No secrets in the repository	The Gemini key and PDI credentials come from environment variables. Git-ignore your .env and commit a .env.example that lists the variable names with empty values.
NFR3	Robustness	A broken payload gets a clear error response. The service never crashes.
NFR4	Reproducible	Anyone can clone your repository, follow your README, and run the service.
NFR5	Honest history	Your commits show how the work actually progressed, not one final bulk commit.

Deliverables

- 1 Your GitHub repository, with the service code and a README that explains step by step how to install and run it.
 - 2 A demo video of 2–4 minutes showing the full loop live: you create a ticket on the PDI, your service processes it, and the decision appears on that same ticket.
 - 3 Screenshots: your Business Rule setup, plus one ticket before and after processing for each of the three decisions (respond, ask, escalate).
 - 4 Your Gemini prompt, committed as a file in the repository (for example prompt.txt or prompts.py), exactly as the service sends it.
 - 5 A reflection file of one page maximum, answering two questions: What was the hardest part? What would you improve with more time?
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